

GREEN BAY STRAUBEL, WI, USA

WMO: 726450

Lat: 44.479N

Lon: 88.137W

Elev: 209

StdP: 98.83

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14898

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.5	-27.5	0.3	-21.8	-24.6	0.4	-19.2	11.5	-7.1	10.4	-7.1	3.7	290	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	31.0	23.1	29.5	22.2	28.1	21.3	24.6	29.2	23.5	27.8	22.5	26.5	4.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.3	27.5	22.1	17.2	26.4	21.1	16.1	25.2	75.6	29.4	71.2	27.8	67.2	26.4	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.1	8.2	-26.2	33.9	3.1	2.0	-28.4	35.4	-30.2	36.6	-32.0	37.7	-34.2	39.2		
(5)			-26.5	26.7	3.0	1.2	-28.6	27.5	-30.4	28.3	-32.1	28.9	-34.3	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-7.8	-6.1	0.0	6.9	13.5	19.1	21.4	20.2	16.2	9.4	2.4	-4.3		
	DBStd	11.35	6.63	6.36	6.21	4.83	4.58	3.91	3.16	2.95	4.29	4.71	5.12	6.09		
	HDD10.0	2199	552	451	315	117	16	0	0	0	4	69	232	442		
	HDD18.3	4177	811	684	568	344	163	37	8	15	90	278	478	701		
	CDD10.0	1342	0	0	6	23	124	273	352	316	191	52	4	1		
	CDD18.3	278	0	0	1	1	13	59	102	72	27	2	0	0		
	CDH23.3	2438	0	0	3	13	153	542	923	551	229	24	0	0		
	CDH26.7	650	0	0	0	1	36	160	282	118	49	3	0	0		
(14) Wind	WSAvg	3.8	4.2	4.3	4.2	4.7	4.1	3.4	3.1	2.8	3.2	3.9	4.1	4.0		
(15) Precipitation	PrecAvg	751	32	27	49	68	78	94	85	86	82	61	51	40		
	PrecMax	998	93	70	119	150	211	261	242	230	198	131	135	145		
	PrecMin	454	3	1	4	12	2	8	21	15	7	9	3	2		
	PrecStd	113	18	16	27	29	42	52	45	45	45	33	30	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	10.6	21.2	25.5	30.1	33.1	33.9	31.5	30.6	26.5	18.8	12.1		
		MCWB	5.0	8.0	15.2	16.9	20.6	23.7	25.5	23.7	22.2	19.2	13.7	10.9		
	2%	DB	4.3	6.3	15.5	21.4	27.3	30.3	31.3	29.4	28.1	22.8	14.6	7.4		
		MCWB	2.5	4.0	11.7	14.4	19.0	22.5	23.7	22.8	21.1	17.5	11.1	5.5		
	5%	DB	2.5	3.9	11.4	18.0	24.7	28.4	29.6	27.9	26.0	20.1	12.3	5.1		
		MCWB	1.4	2.2	8.0	11.8	17.6	21.2	22.5	21.7	19.9	15.6	9.2	3.8		
	10%	DB	1.2	2.3	8.2	15.3	22.0	26.6	28.0	26.4	24.0	17.3	9.9	3.2		
		MCWB	0.4	0.8	5.3	9.9	16.0	20.1	21.5	20.6	18.8	14.0	7.5	1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	8.2	16.1	17.9	22.2	25.5	26.8	25.3	23.9	21.1	14.5	11.1		
		MCDB	7.2	10.6	20.9	23.2	27.5	30.9	32.4	29.4	28.4	24.7	17.6	12.1		
	2%	WB	2.8	4.1	12.1	15.2	20.3	23.8	25.0	24.0	22.3	18.5	11.9	6.1		
		MCDB	3.8	5.6	14.9	20.2	25.3	29.0	29.8	28.0	26.3	21.6	14.2	7.0		
	5%	WB	1.6	2.4	8.2	12.6	18.6	22.4	23.8	22.9	21.0	16.3	9.7	3.9		
		MCDB	2.4	3.8	10.9	17.3	23.7	27.0	28.0	26.3	24.6	19.6	11.9	5.0		
	10%	WB	0.4	0.9	5.4	10.3	16.8	21.0	22.6	21.8	19.8	14.2	7.7	1.9		
		MCDB	1.3	2.2	8.2	14.5	21.4	25.2	26.7	25.1	23.0	17.1	9.9	2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.4	9.0	10.3	11.2	10.8	11.3	10.9	11.6	9.9	7.9	7.1		
		MCDBR	7.7	8.3	12.5	15.6	14.5	12.7	12.5	12.2	13.5	13.4	11.5	8.4		
		MCWBR	6.9	6.7	8.8	9.7	7.9	6.6	6.4	6.5	7.4	8.5	8.5	7.1		
		MCDBR	7.3	7.8	12.2	14.6	13.0	11.4	11.2	10.4	11.6	12.0	10.6	7.8		
	5% WB	MCWBR	6.7	6.6	9.1	9.7	7.6	6.8	6.5	6.2	7.0	8.7	8.5	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.320	0.343	0.376	0.406	0.412	0.398	0.398	0.371	0.330	0.300	0.275		
	taud		2.410	2.241	2.326	2.296	2.258	2.280	2.325	2.318	2.382	2.495	2.506	2.449		
	Ebn at Noon		864	857	889	885	865	859	867	853	849	847	820	820		
	Edh at Noon		73	104	110	124	133	130	123	120	104	81	67	64		
(44) All-Sky Solar Radiation	RadAvg		1.57	2.62	3.78	4.52	5.47	5.95	6.11	5.21	4.12	2.60	1.64	1.18		
	RadStd		0.18	0.25	0.38	0.46	0.44	0.42	0.27	0.34	0.32	0.26	0.15	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page